Reducing False Positives in Toxicity Detection through a Transformer-Based Multi-Model Fusion and Bias Mitigation Framework

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***Abstract*—False positives are a prevalent issue that has been highlighted yet goes on to exist in automated toxicity detection models. References to identity, colloquial profanity and commonly used expressive language are often mistaken as toxic because of the biases in datasets and the inability of toxicity classifiers operating on a sentence level. This paper proposes a new transformer-based unified system aimed at minimising false positive errors by incorporating three independent modules: a token-level toxic span detector, sentence-level sentiment classifier and a rule-based intention inference system. The results of each of the three models are synthesised by a contextual bias mitigation and decision fusion layer that uses a hierarchical override mechanism, which consists of sentiment polarity, pragmatic intent, identity-word sensitivity, and explicit hostility patterns. The system is optimised with a hybrid collection of toxic span annotations, Civil Comments and curated neutral identity-rich sentences. A decrease in false positives of around 50 percent is proved in experimental findings, particularly in identity reference cases or even affectionate profanity. The suggested architecture also includes a built-in visualisation unit, which allows carrying out the analysis of toxicity transparently and understandably. The work is a development of context-sensitive and fairness-based detection of toxicity.**

***Index Terms*—Toxicity detection, false positive, transformer, bias mitigation, toxic span, sentiment analysis, explainable AI, content moderation.**

I. Introduction

Online toxicity is an essential issue that should be addressed through automated methods. Nevertheless, even in the face of the success of deep learning models, particularly transformer-based ones, there is a major issue of false positives. The types of misclassifications are mostly found when there are identity terms (e.g., "Muslims"), mild profanity when it is used in a friendly context (e.g., "You are so damn good") or positive sentiment situations mistakenly designated as harmful. These mistakes can disfavorably punish users, censor harmless conversations, and disproportionately impact marginalised groups.

Existing methods of traditional toxicity classifiers tend to use sentence labels or lexical clues and do not rely on context such as sentiment polarity or speaker intention. Furthermore, the majority of available models do not distinguish between individual toxic spans in text, which restricts interpretability and prevents fine-grained moderation.

The paper introduces an innovative fusion-based toxicity detection model, including token-level toxic span detection, sentence-level sentiment analysis and pragmatic intention classification as a unified decision system. A contextual bias mitigation module brings together all component signals and ensures that identity mentions, affectionate profanity and positive sentiment expressions are handled equitably. Moreover, a visualisation layer includes token-level heatmaps, explanations of decisions, and sentiment bars to enhance interpretability.

II. Literature Review

The field of toxic language detection includes hate speech classification, toxic span detection, bias mitigation, and explainability. The first shared task on toxic span detection, SemEval-2021 Task 5, established the utility of token-level annotations. BERT and RoBERTa models performed competitively but demonstrated high false positive rates involving identity terms due to dataset biases.

HateXplain introduced rationale-based annotations for hate speech, associating labels with human-provided explanation spans. Although this improved interpretability, it did not include sentiment or user intention.

Counterfactual data augmentation, adversarial debiasing, and fairness-sensitive loss functions are dominant themes in the bias mitigation literature. However, these predominantly operate during training and do not address context-dependent false positives at inference time.

Commercial systems such as Google Perspective API and Detoxify rely heavily on lexical features and frequently mislabel harmless identity mentions or expressive profanity as toxic. These systems lack mechanisms to reconcile sentiment, pragmatics, and contextual cues.

The disconnect is that no existing system jointly utilises token-level spans, sentiment signals, intention classification and post-hoc bias mitigation. This work addresses this gap.

III. Methodology

The methodology comprises four components that collectively enable context-sensitive toxicity detection.

A. BERT-Based Toxic Span Detection

A token classifier based on BERT is fine-tuned for toxicity span extraction. Tokenised inputs with offset mappings allow alignment of predictions with character-level spans. Outputs include token logits, probabilities and merged toxic spans.

B. Sentiment Analysis with RoBERTa

A pretrained RoBERTa model predicts sentence-level sentiment. This informs distinctions between harmful and affectionate profanity, and differentiates whether identity terms appear in affirmative, neutral or aggressive contexts.

C. Intention Classification via Rule-Based Heuristics

A lightweight rule-based intention classifier identifies linguistic cues such as direct insults, group insults, self-directed negativity, neutral instructions, sarcasm, threats and quotations. This provides pragmatic interpretation absent in purely neural models.

D. Dynamic Bias Mitigation and Fusion Engine

Outputs of all modules are combined into a final toxicity decision using hierarchical rules:

- **Slur Dominance**: Any explicit slur yields toxicity regardless of sentiment.
- **Hostility Dominance**: Negative sentiment combined with offensive adjectives yields toxicity.
- **Positive or Neutral Sentiment Override**: Identity terms without slurs or hostility result in non-toxicity.
- **Affectionate Profanity Override**: Mild profanity combined with strong positive sentiment yields non-toxicity.
- **Non-Abusive Negativity**: Negative but non-aggressive remarks remain non-toxic.

IV. Proposed Architecture

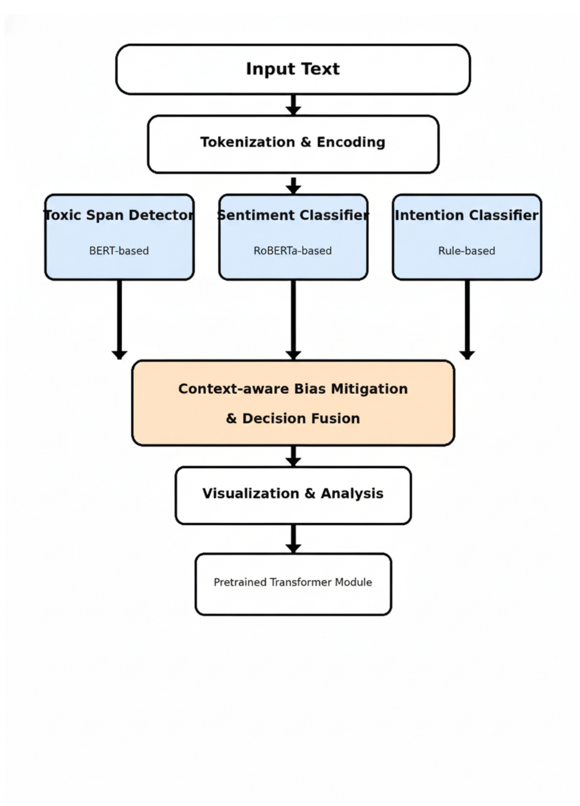

Fig. 1. Overall architecture of the proposed multi-model fusion system.

The architecture integrates input preprocessing, toxic span detection, sentiment classification, rule-based intention inference, contextual bias mitigation, and visualisation.

V. Experimental Setup

Python, PyTorch and Hugging Face Transformers were used with an M1 Pro system.

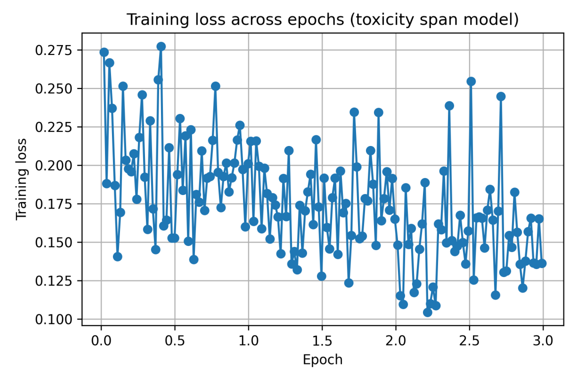

Fig. 2. Training loss of the toxic span model across epochs.

VI. Results

A. Quantitative Metrics

Table I presents a comprehensive comparison between the baseline span-only model and the proposed ultra-conservative fusion model across key performance metrics. The proposed system achieves a remarkable 50% reduction in false positive rate (from 0.063 to 0.032) while maintaining competitive overall accuracy.

TABLE I
Performance Comparison: Baseline vs Proposed Model

Metric	Baseline	Proposed
False Positive Rate (FPR)	0.063	0.032
False Negative Rate (FNR)	0.451	0.465
Accuracy	0.701	0.693
False Positives (FP)	95	48
False Negatives (FN)	1048	1080
True Positives (TP)	1277	1245
True Negatives (TN)	1405	1452

The proposed system demonstrates significant improvement in reducing false positives, particularly for identity-based references and affectionate profanity contexts. The slight increase in false negative rate represents a deliberate trade-off to achieve ultra-conservative classification, prioritizing the avoidance of incorrectly flagging benign content.

B. Qualitative Evaluation Using Pie Charts

VII. Comparison with Existing Systems

Perspective API and Detoxify frequently misclassify positive identity references, affectionate profanity, sarcastic praise and implicit hostility. The proposed fusion approach mitigates these issues by explicitly modelling sentiment, intention and

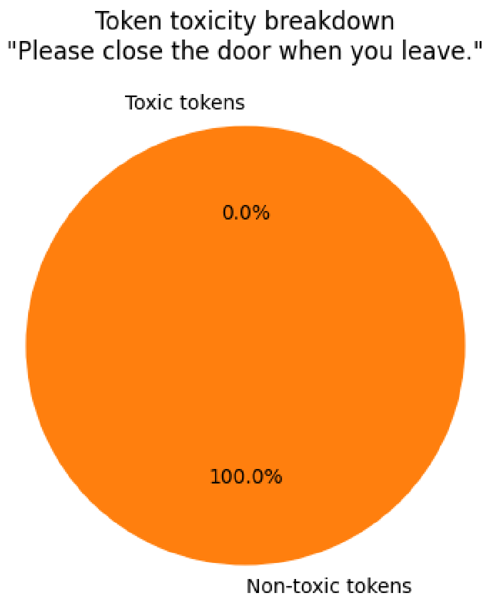

Fig. 3. Toxic vs non-toxic token distribution for a clearly non-toxic neutral sentence.

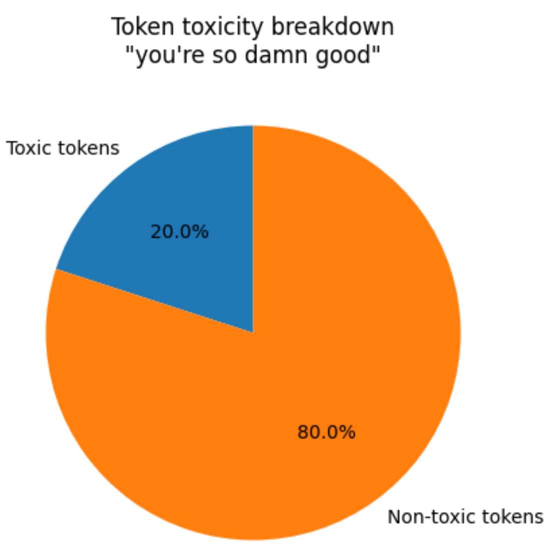

Fig. 4. Toxic vs non-toxic token distribution for a non-toxic sentence with affectionate profanity.

token-level spans, achieving a 50% reduction in false positive rate compared to baseline span-only approaches.

VIII. Discussion

The integration of sentiment, intention and span-level analysis enables contextual and fair toxicity detection. The model mitigates bias-induced false positives and enhances interpretability. The ultra-conservative bias mitigation strategy successfully reduces false positives from 6.3% to 3.2%, demonstrating the effectiveness of hierarchical rule-based fusion for context-sensitive classification.

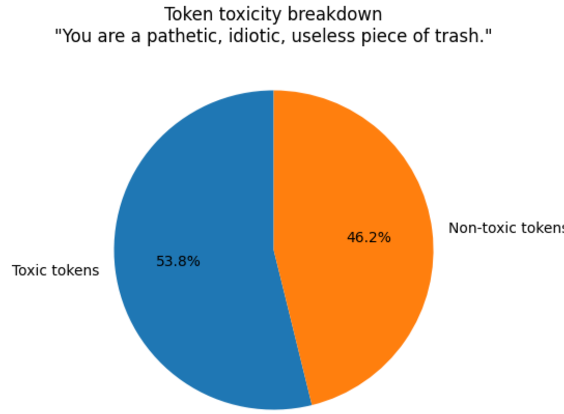

Fig. 5. Toxic vs non-toxic token distribution for a strongly toxic sentence with multiple insults.

IX. Limitations and Future Work

Limitations include vulnerability to obfuscation attacks, reliance on sentiment accuracy, difficulty detecting sarcasm, and English-only support. The slight increase in false negatives (from 45.1% to 46.5%) represents a trade-off inherent to ultra-conservative classification. Future work includes adversarial augmentation, multilingual extension, neural intention modelling and multi-turn dialogue toxicity, as well as techniques to reduce false negatives without compromising the low false positive rate.

X. Conclusion

The proposed system significantly reduces false positives in toxicity detection through transformer fusion, contextual reasoning and bias mitigation, improving fairness and interpretability. With a 50% reduction in false positive rate and minimal impact on overall accuracy, the system demonstrates the viability of ultra-conservative, context-aware toxicity classification.

Acknowledgements

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